JEE-Main-06-04-2024 (Memory Based)
[EVENING SHIFT]

Physics

Question: Energy associated with 10 helium gas atoms at temperature 'T' is
(where KB = Boltzmann constant, R = universal gas constant)
Options:
(a) 15 RT
(b) 30 RT
(c) 30 K_BT
(d) 15 K_BT
Answer: (d)

Question: An object of weight 200 N is suspended through a chain of mass 10 kg from a branch of a tree. Calculate the force on chain applied by branch of tree
Options:
(a) 200 N
(b) 300 N
(c) 100 N
(d) 210 N
Answer: (b)

Question: While finding the refractive index of a glass slab, the travelling microscope is focused on ink dot on a white paper. When a glass slab of thickness 5.1 cm is placed on this inK dot, the microscope is raised through 1.7 cm. Then refractive index of glass slab is
Options:
(a) 3/2
(b) 4/3
(c) 2/3
(d) 2
Answer: (a)

Question: A p-type semiconductor has acceptor levels 6eV above the valence band. The maximum wavelength of light required to create a hole is
(Planck's constant , h = 6.6 × 10^{-34} J-s)
Options:
(a) 2060 nm
(b) 206 nm
(c) 206 Å
(d) 20.6 nm
Answer: (b)

Question: 2 open organ pipes one of 60 cm and one of 90 cm are in their 6th and 5th harmonic frequencies respectively, velocity of sound is 333 m/s find the difference in their frequencies
Options:
(a) 247 Hz
(b) 200 Hz
(c) 70 Hz
(d) 100 Hz
Answer: (a)

Question: The electric field in an electromagnetic wave is given by **E = 600** sin (**ωt** - kx)**N/C**. Then the intensity of the wave if it is propagating along x-axis in free space (Gien ε_0 **= 9 × 10^{-12} C^2 N^{-1} m^{-2})**
Options:
(a) 972 w/m^2
(b) 243 w/m^2
(c) 648 w/m^2
(d) 486 w/m^2
Answer: (d)

Question: Two spheres each of charge q, repel each other by 16 N. A Third identical, uncharged sphere touched by these both spheres one after another. The new force between the spheres is
Options:
(a) 16 N
(b) 6 N
(c) 2 N
(d) 20 N
Answer: (b)

Question: If threshold energy required to eject an electron from emitter is 10.56 × 10^{-20} J, then corresponding maximum value of wavelength is
Options:
(a) 1.875 × 10^{-7} m
(b) 1.875 × 10^{-6} m
(c) 187.5 Å
(d) 187.5 nm
Answer: (b)

Question: If 48 J heat is given to one mole of helium gas and increase in temperature is 2° C. Work done is (R = 8.3 J K^{-1} mol^{-1})
Options:
(a) 12.5 J
(b) 23.1 J
(c) 25.2 J
(d) 48 J
Answer: (b)

Question: Energy of a photon is 2.5 eV and work function is 1.5eV. Find the stopping potential
Options:
(a) 2V
(b) 3V
(c) 1V
(d) 4V
Answer: (c)

Question: Current In an inductor from -2A to +2A in 0.2 second generating an induced emf of 1 volt. Find self inductance
Options:
(a) 5 mH
(b) 50 mH
(c) 10 mH
(d) 100 mH
Answer: (b)

Question: Weight of an object on the surface of the earth is 300 N. Find the weight at a depth of R/4
Options:
(a) 150 N
(b) 300 N
(c) 100 N
(d) 225 N
Answer: (d)

Question: Kinetic energy is increased 36 times of its initial Value find Percentage increase in momentum
Options:
(a) 100%
(b) 500%
(c) 1000%
(d) 700%
Answer: (b)

Question: Statement-1: Dimensional formula of specific heat capacity is $L^2 T^{-2} K^{-1}$
Statement-2: Dimensional formula of gas constant is $M^1L^{-1}T^{-2}K^{-1}$

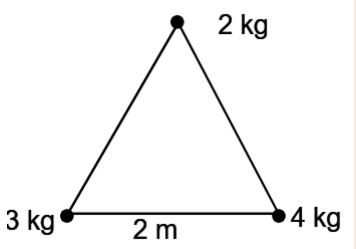
Options:
(a) Only statement 1 is correct
(b) Only statement 2 is correct
(c) Both are correct
(d) Both are incorrect
Answer: (a)

Question: A heat of 42 J was given to 1 mole of helium gas in a sealed tank. Find work done by the gas.
(Given R = 8.31 J/mol^{-k})
Options:
(a) 42 J

(b) Zero
(c) 84 J
(d) 21 J
Answer: (b)

Question: A ball is thrown vertically upwards and take time t_1. And if thrown vertically downward then take time t_2. Then if ball is just dropped how much time will take?
Options:
(a) $\sqrt{t_1} + t_2$
(b) $\sqrt{t_1} - t_2$
(c) $\sqrt{t_1}/t_2$
(d) $\sqrt{t_1 t_2}$
Answer: (d)

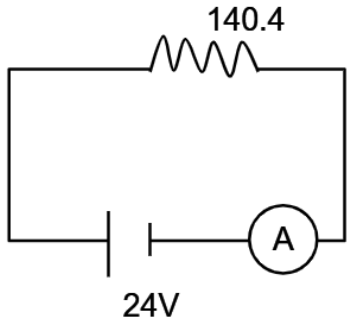
Question: Three point masses 2kg, 3kg and 4kg are placed at the vertices of an equilateral triangle. Find moment of inertia about centroid of triangle perpendicular to the plane.



Options:
(a) 12 kg-m^2
(b) 18 kg-m^2
(c) 9 kg-m^2
(d) 36 kg-m^2
Answer: (a)

Question: Pressure Difference between inside & outside the bubble is
Options:
(a) 2S/r
(b) S/r
(c) 4S/r
(d) 4Sr
Answer: (c)

Question: Ammeter A has coil of resistance 240 ohm and is shunted with resistance 10 ohm. Find reading of A (in mA)



Options:
(a) 100 mA
(b) 120 mA
(c) 140 mA
(d) 160 mA
Answer: (d)

Question: Focal length for a thin convex lens is 20 cm whose radii of curvature are 15 cm & 30 cm. Determine refractive index of lens
Options:
(a) 1.2
(b) 1.3
(c) 1.4
(d) 1.5
Answer: (d)

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Chemistry

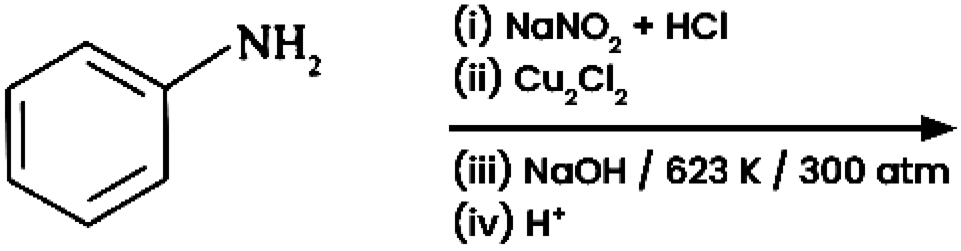
Question: Calculate molality of 3M NaCl solution. Density = 1.25 g/ml
Options:
Answer: (3 Molal)

Question: Consider given reaction, identify major product P

$$CH_3COOH \xrightarrow[\text{(iv) } H_2O/OH^-/\Delta]{\text{(i) } LiAlH_4 \text{ (ii) PCC (iii) } HCN/OH^-}$$

Options:
Answer: ()

Question:

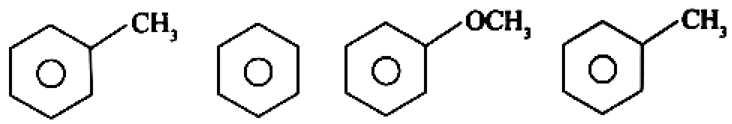


Options:
(a)
(b)
(c)
(d)
Answer: ()

Question: Identify the product formed in the following reaction:

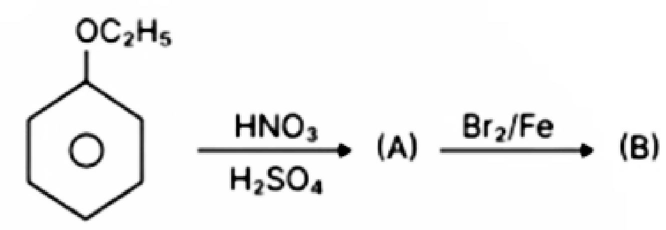
Options:
Answer: ()

Question: Decreasing order of electrophilic Substitution

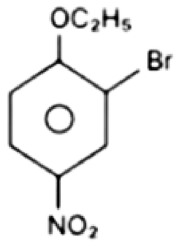


Options:
Answer: ()

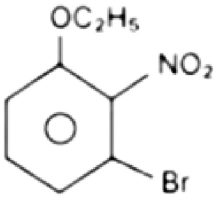
Question: Product B is :



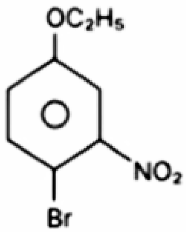
Options:



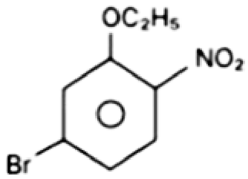
(a)



(b)



(c)



(d)

Answer: ()

Question: For a Certain Reaction, dH = 400 Kj/mol, dS = 200 J/mol K. At what minimum temperature will this process be spontaneous?

Options:

Answer: ()

Question: The Shortest wavelength of Paschenn Series for H atom is?

Options:

(a) 9/R
(b) 16/R
(c) 144/R
(d) 7R/144
Answer: ()

Question: Which of the following d-block elements has maximum unpaired electron in ground state electronic configuration ?
Options:
(a) Ti (22)
(b) V(23)
(c) MN(25)
(d) Cr(24)
Answer: ()

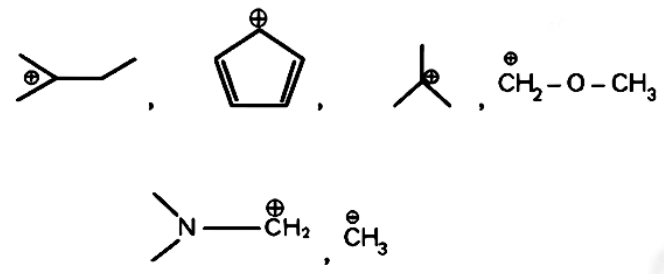
Question: How do you convert a Galvanic Cell into an Electrolytic Cell
Options:
(a) By interchanging Cathode and Anode
(b) By Applying External Voltage lesser than Cell EMF
(c) By Applying External Voltage more than Cell EMF
(d) By applying internal voltage
Answer: ()

Question: Select incorrect statement :
Options:
(a) Enzymes are bio catalysts
(b) Enzymes are nonspecific and catalyze most type of reactions
(c) Most of the enzymes are globular proteins
(d) Enzyme oxidase catalyses conversion of maltose to glucose.
Answer: ()

Question: spin only magnetic moment for the oxidizing agent among
VO^{2+} MnO^-_4 $Cr_2O^{2-}_7$
Options:
Answer: ()

Question: Sp^2 :- NH_3 SO_2 , SiO_2 , C_2H_2 , CO_2 $Becl_2$, C_2H_4 , C_6H_6 , $C_2H_4Cl_2$
Options:
Answer: ()

Question: Number of Carbocation which is not Stabilized by Hyperconjugation



Options:
(a)
(b)
(c)
(d)
Answer: ()

Question: IUPAC name of [PtBr2 (PMe3)2]
Options:
(a) Dibromido(trimethylphosphine) platinum(II)
(b) Dibromido(trimethylphosphine) platinum(IV)
(c) Dibromido(trimethylphosphine) platinate (IV)
(d) (trimethylphosphine) Dibromidoplatinum (IV)
Answer: ()

Question:

$$\text{Salt} \xrightarrow{AgNO_3} \text{yellow ppt} \xrightarrow{NH_4OH} \text{insoluble}$$

Salt may contain anion -

Options:
(a) I^-
(b) Br^-
(c) Cl^-
(d) F^-
Answer: ()

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Maths

Question: If the words with or without meaning made using all the letters of the word 'NAGPUR' are arranged in dictionary order. Then the word 315th position

Options:

(a) NRAPUG

(b) NRAGUP

(c) NRAGPU

(d) NRAPGU

Answer: (d)

Question: ${}^{n+1}C_{r+1} : {}^{n}C_{r} : {}^{n-1}C_{r-1} = 55 : 35 : 21$ then $2n + 5r$ is equal to

Options:

(a) 60

(b) 55

(c) 62

(d) 50

Answer: (d)

Question: There are letters to be delivered to 5 different location, then find the probability that letter is delivered to exactly 2 correct assuming each letter is delivered to unique address.

Options:

(a) 18/25

(b) 12/25

(c) 6/25

(d) 4/25

Answer: (c)

Question: $f(x) = \frac{1}{7-\sin 5x}$ be a function defined on R.

Then the range of the function f(x) is equal to:

Options:

(a) $\left[\frac{1}{7}, 1\frac{1}{6}\right]$

(b) $\left[\frac{1}{8}, \frac{1}{6}\right]$

(c) $\left[\frac{1}{8}, \frac{1}{5}\right]$

(d) $\left[\frac{1}{7}, \frac{1}{5}\right]$

Answer: (b)

Question: If $\int_{o}^{\frac{\pi}{2}} \frac{dx}{a^2\sin^2 x + b^2\cos^2 x} = \frac{1}{12}\tan^{-1}(3\tan x) + c,$ then the maximum value of a sin x + b cosx is :

Options:

(a) $\sqrt{10}$

(b) $\sqrt{20}$

(c) $2\sqrt{10}$

(d) $2\sqrt{5}$

Answer: (c)

Question: If the function $f(x) = \left(\frac{1}{x}\right)^{2x} x > 0,$ attains the maximum value at x = 1/e then

Options:

(a) $e^{\pi} < \pi^{e}$

(b) $e^{2\pi} < (2\pi)^{e}$

(c) $(2e)^{\pi} > (\pi)^{2e}$

(d) $e^{\pi} > \pi^{e}$

Answer: (d)

Question: α, β are the roots $x^2 + \sqrt{2}x - 8 = 0$. Let $U_n = \alpha_n + \beta_n$ then $\frac{U_{10}+\sqrt{2}U_9}{2U_8}$ is equal to ____

Answer: (4)

Question: If the area bounded by the region (c, y) such that

$\left\{(x, y) \middle| \frac{a}{x^2} < y < \frac{1}{x};\ 1 < x < 2,\ 0 < a < 1\right\}$

is $\left(\ln\ 2 - \frac{2}{7}\right)$ Sq. units, then (7a - 3) is equal to:

Answer: (1)

Question: If $\vec{a} = 2i - j + k$ and $\vec{b} = \left(\left(\vec{a} \times (i + j) \times i\right) \times i\right),$

then the square of projection of $\vec{a}$ on $\vec{b}$

Answer: ()

Question: $\lim_{n\to\infty} \frac{\sum\left(n^4 - 2n^3 + n^2\right)}{\sum\left((3n)^4 + n^3 - n^2\right)}$ is equal to

Options:

(a) 1/81

(b) 1/72
(c) 1/57
(d) 1/93
Answer: (a)

Question: If the order of matrix A is 3 and |a| = 3 then the value of det(adj(-4adj(-3adj(2A-1)))) is $2^m 3^n$. The value of
m + 2n is:
Answer: (32)

Question: If $\int_0^3 \left([x^2] + \left[\frac{x^2}{2}\right]\right)dx = a + b\sqrt{2} + c\sqrt{6} - \sqrt{3} - \sqrt{5} - \sqrt{7}$ (a, b, c, ∈, $\mathbb{I}$), then (a + b + c) equal to ______ .

Answer: ()

Question:If (α,β,γ) is the mirror image of Q(3, -3, 1) in the line $\frac{x-0}{1} = \frac{y-3}{1} = \frac{z-5}{-1}$ and R(2,5,3). If the area of ΔPQR is λ. The $\frac{\lambda^2}{546}$ equals to :
Answer: ()

Question: Sides of a triangle are AB = 9, BC = 7, AC =8. Find cos 3C.
Answer: ()